Hierarchical MARL with Cyclic Braking for Cooperative Control of Heavy-Haul Train Platoons on Long Downhill Sections

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Abstract—Heavy-haul trains operating on long downhill sections require cyclic air braking, which produces large in-train longitudinal (coupler) forces. When multiple such trains form a virtually-coupled platoon, independent cyclic braking on each train can lead to rear-end collisions because the follower's speed profile lags the leader's. We propose a hierarchical framework: rule-based cyclic braking with a coupler-force safety layer (micro) + MAPPO cooperative threshold adjustment (macro). On the real Shuozhou–Huanghua profile with 10,000-tonne trains, our cooperative policy (M1) never collides across randomized initial conditions and under increasing pressure, whereas independent control (M3), a rule-based cooperative heuristic (M4), and a naive DQN baseline (M-DQN) all collide in a subset of conditions. The coupler-force constraint ($\leq$1000 kN) is enforced by a hard safety layer inside the rule-based cyclic-braking controller and is satisfied in every method that includes it; the naive DQN, which lacks both this safety layer and inter-train cooperation, violates the limit. The cooperative gain grows monotonically with scenario pressure, and the benefit scales to 5-train platoons.

Index Terms—Heavy-haul train, virtual coupling, multi-agent reinforcement learning, cyclic braking, coupler force, long downhill.

I. Introduction

Heavy-haul trains operating on long downhill sections (e.g., the Shuozhou–Huanghua line, 23.8 km with sustained -1.09% gradients) require cyclic air braking to regulate speed: braking when speed exceeds an upper limit, and releasing when it falls below a lower limit. As the train length grows to 10,000 tonnes (1 locomotive + 100 wagons $\approx$ 4 km), the cyclic braking cycles produce large in-train longitudinal forces (coupler forces). Exceeding the coupler-force safety limit (1000 kN in normal operation, per the China Academy of Railway Sciences) risks coupler failure.

When multiple such heavy-haul trains operate as a virtually-coupled platoon (to increase line capacity), each train must simultaneously (i) regulate its own speed via cyclic braking and (ii) maintain a safe inter-train spacing to avoid rear-end collision. Independent control of each train is fragile: under a long downhill profile, the follower's cyclic-braking speed profile can lag the leader's, and the inter-train gap collapses into a rear-end collision in a subset of operating conditions.

Prior work falls into three gaps:

- Long-downhill RL for a single train (Zhang et al. 2024 DQN; Liu et al. 2021; Wang et al. 2025) explicitly ignores coupler forces (single-point mass).
- Virtually-coupled MARL (Zhang et al. 2021; Guo et al. 2025) targets urban/metro or car platoons and ignores coupler forces.
- Coupler-force optimization (Wei 2024; Fu 2019) is single-train and uses optimization (QP/GA), not learning.

Our contribution: a hierarchical framework combining rule-based cyclic braking with a coupler-force safety layer (micro) and RL-based cooperative threshold adjustment (macro), for cooperative control of heavy-haul train platoons on long downhill sections, which

1) guarantees coupler-force safety ($\leq$ 1000 kN): the coupler-force constraint is enforced inside the intra-train rule-based cyclic-braking controller (a hard safety layer), so it is satisfied regardless of how the cooperative layer is designed — an ablation (M1 vs. a variant ignoring coupler forces) shows identical coupler-force and safety performance, confirming the constraint is robustly enforced at the physical-control layer rather than left to the learned policy;
2) robustly prevents rear-end collision: the RL cooperative policy (M1) never collides across all randomized initial conditions (0 ± 0 collisions), whereas independent control (M3), a rule-based cooperative heuristic (M4), and a naive DQN baseline that lacks both the coupler-force safety layer and inter-train cooperation (M-DQN) collide in a subset of conditions (45.7 ± 80.7, 45.1 ± 80.8, and 1434 collision steps respectively). A fair DQN with the same architecture, safety layer, observation, and reward as M1 but with DQN instead of MAPPO for cooperative adjustment still collides on average (89.2 ± 109.4), isolating the benefit to the MAPPO algorithm itself rather than to the hierarchical architecture;
3) scales to larger platoons: the cooperative gain is stable across 3- and 5-train platoons (+67,597 / +67,595), and grows monotonically under stronger

pressure (+67,597 → +89,588 as the initial gap shrinks from 100 to 40 m).

II. Related Work

Cyclic air braking RL (single train, no coupler force). Zhang et al. [1] propose a DQN for cyclic air braking on the Shuozhou–Huanghua line with speed range $[40, 80]$ km/h, but explicitly state "interaction forces between the cars are not taken into account." Wang et al. [2] use DQN for longitudinal-impulse suppression with simplified dynamics. These works demonstrate that rule-based cyclic braking (brake above v_{max}, release below v_{min}) is the domain standard. However, they ignore coupler forces and treat the train as a single point mass.

Virtually-coupled MARL. Zhang et al. [3] use multi-agent deep RL for train-platoon consistency with an artificial potential field, on a flat metro line with no slope and no coupler forces. Guo et al. [4] propose a MARL consensus controller for autonomous platoons. These works target urban/car scenarios, ignore coupler forces, and assume free ±acceleration, which does not hold on long downhill sections.

Coupler-force optimization and MPC. Wei et al. [5] propose a hierarchical framework (QP top layer + longitudinal dynamics bottom layer) to enforce coupler-force constraints for a single train. Fu et al. [6] use genetic algorithm + generalized predictive control for single-train operation under coupler constraints. Most directly, Yang et al. [8] use model predictive control (MPC) for single heavy-haul trains on the same Shuozhou–Huanghua long-downhill profile with cyclic braking and coupler-force impulse minimization. These are single-train, model-based approaches; our work extends to multi-train virtual-coupling with model-free MARL, which removes the need for an explicit coupled multi-train prediction model.

Our position. We fill the cross-blank: multi-train virtual coupling + long-downhill cyclic braking + explicit coupler-force modeling, using a hierarchical framework where rule-based cyclic braking (the domain standard, equipped with a hard coupler-force safety layer) handles micro speed control, and RL (MAPPO) handles macro cooperative threshold adjustment. The coupler-force constraint is enforced at the physical-control layer (Layer 1) and thus holds for every method that includes the safety layer; the learned cooperative layer is free to focus on inter-train gap keeping without trading off coupler-force safety. We demonstrate that the RL cooperative policy provides robust rear-end-collision prevention (never collides across initial conditions), which is not achieved by independent control, a rule-based heuristic, or a naive DQN.

III. Problem and System Model

A. Longitudinal train dynamics with air-brake release wave

Each train is modeled as a multi-body longitudinal dynamics system (following NeTrainSim). For wagon i with mass m_i, position x_i, and velocity v_i:

$$m_i \ddot{x}_i = F_{t,i} - F_{r,i} - F_{b,i} - F_{g,i} + F_{c,i-1} - F_{c,i}, \qquad (1)$$

where $F_{t,i}$ is tractive force, $F_{r,i}$ is Davis resistance, $F_{b,i}$ is braking force (electric on locomotive + air brake propagating along the consist at finite brake-wave speed), $F_{g,i}$ is grade resistance, and $F_{c,i}$ is the coupler force between wagons i and i+1, modeled as a spring–damper with dead zone (coupler gap):

$$F_c = \begin{cases} k(e - \text{gap}) + c\dot{e}, & |e| > \text{gap}, \\ c\dot{e}, & |e| \le \text{gap}, \end{cases} \qquad (2)$$

with $e = x_i - x_{i+1} - l_0$. The coupler-force safety limit is 1000 kN [6].

Air-brake release wave (physical modeling). When the brake is released, the release command propagates from the lead car backward at a finite speed (∼200 m/s) as a continuous wavefront. We model this correctly: each wagon's release delay is d_i/wave_speed (where d_i is the distance from the brake source), and all wagons release in parallel as the wavefront reaches them. This yields a physically correct release time of ∼6 s for a 1221 m consist (verified: 61 steps vs theoretical 61 steps, ratio 1.0×), and preserves the "maximum coupler force occurs after release" phenomenon noted in the literature [7].

B. Train platoon scenario

Three 10,000-tonne trains (1 locomotive + 100 wagons each, total ≈ 4 km) operate on the Shuozhou–Huanghua profile shuozhou_23800 with speed limits $[40, 80]$ km/h. The target inter-train gap is 100 m (headway 5 s). Trains start on the same grade section at initial speed 65 km/h. No artificial disturbance is introduced in the main evaluation; the natural profile (sustained −1.09%) drives cyclic braking.

IV. Proposed Method

A. Hierarchical architecture

We use a two-layer structure (Fig. 1):

- Layer 1 (intra-train, Coupler Level) — rule-based cyclic braking. Each train uses get_action_cyclic_braking: brake at throttle $u = -0.75$ when speed $v \ge v_{max}$, release when $v \le v_{min}$ (hysteresis). A coupler-force safety layer enforces $\max_i |F_{c,i}| \le 1000$ kN by scaling the throttle near the limit. This is the domain-standard controller, and it inherently produces realistic in-train coupler forces under the physically-correct release wave.
- Layer 2 (inter-train, Platoon Level) — MAPPO cooperative threshold adjustment. Each train is an agent. The agent outputs a cooperative adjustment $c_{\text{adj}} \in [-1, 1]$, mapped to threshold offsets $v_{max}^{adj}, v_{min}^{adj} \in [\pm 5\,\text{km/h}]$ that shift the cyclic-braking thresholds of Layer 1. The adjustment does not directly manipulate

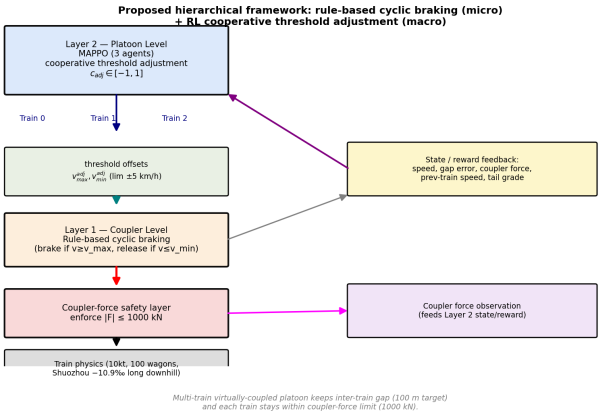


Fig. 1. Hierarchical framework: rule-based cyclic braking (micro) + RL cooperative threshold adjustment (macro).

TABLE I
Experimental settings.

Parameter	Value
Consist preset	10,000 t, 1 loco + 100 wagons (≈4 km)
Profile	Shuozhou–Huanghua, sustained −10.9‰
Speed limits	[40, 80] km/h (release / line)
Target inter-train gap	100 m (headway 5 s)
Train horizon / eval horizon	300 / 3000 steps
MAPPO episodes / lr / entropy	200 / 5×10^{-5} / 0.001
adj_coef	0.3
Dynamics dt	0.1 s
Coupler limit	1000 / 2250 kN
Brake-wave speed	200 m/s
# train seeds / eval seeds	3 / 10

the throttle (avoiding exploration that breaks rule-based cyclic braking). A positive adjustment makes the train brake earlier / release later.

B. MDP formulation

- State (per agent, 11-dim): normalized own speed error, gap error, coupler-force statistic, position, tail grade, previous-train speed, previous-train braking indicator and deceleration strength.
- Action: $c_{\text{adj}} \in [-1, 1]$ (cooperative threshold offset).
- Reward: speed-band penalty + gap-keeping reward + speed-consistency + rear-end potential-field penalty + coupler-force penalty.

C. Training and experimental settings

We use MAPPO (shared policy + centralized critic + GAE + KL early-stopping), 3 trains, 10,000-tonne preset, Shuozhou profile, 200 episodes, adj_coef = 0.3, entropy = 0.001. Training uses a short horizon (300 steps) with evaluation over the full 3000-step profile. Table I summarizes the experimental settings.

V. Experiments

A. Evaluation protocol and baselines

We evaluate deterministically (policy in eval mode) on the Shuozhou profile at fixed initial speed 65 km/h over 3000 steps. For robustness statistics, we randomize the initial speed in $[58, 66]$ km/h over 10 seeds and report mean±std. We evaluate on both 3-train and 5-train platoons.

Methods:

- M1 (proposed): Layer-1 rule cyclic braking + Layer-2 MAPPO cooperative threshold adjustment.
- M3 (independent): Layer-1 rule cyclic braking only, δ=0 (no cooperation). Same architecture.
- M4 (rule-based cooperative heuristic): Layer-1 rule cyclic braking + a hand-crafted PID-like rule mapping gap error and speed difference to c_{adj}. Same architecture; only the c_{adj} source differs (rule vs RL).
- M-DQN: a naive DQN baseline [1] trained to learn cyclic braking; in the multi-train scenario each train runs an independent DQN (no inter-train cooperation) and without the Layer-1 coupler-force safety layer (aligned with [1], which ignores coupler forces). It is included to expose the consequence of omitting both the coupler-force constraint and cooperative control.
- M-DQN-fair: a fair DQN baseline that uses the same hierarchical architecture as M1 (Layer-1 rule cyclic braking + coupler-force safety layer) but replaces the Layer-2 MAPPO with a per-train DQN outputting discrete cooperative threshold offsets. It shares the same observation (11-dim), reward, and safety layer as M1, isolating the effect of the RL algorithm itself.
- Coast: throttle = 0 baseline.

TABLE II
Main result (3-train platoon, fixed init 65 km/h).

Metric	Coast	M1	M3	M4	M-DQN-fair	M-DQN
Reward	+27,950	+36,446	−31,151	−33,991	−38,432	−606,185
Overspeed (%)	5.3	1.3	0.8	0.8	0.8	—
Under-speed (%)	0.0	0.9	4.5	4.5	4.7	—
Peak coupler (kN)	651	804	734	734	900	2,111
Min gap (m)	182	182	−321	−321	−347	−234
Collision steps	0	0	202	202	229	1,434

B. Main result (fixed initial speed 65 km/h, 3-train)

Table II summarizes the main result. M1 avoids rear-end collisions (0 vs 202), keeping the min gap at 182 m, while M3/M4's independent control leads to gap collapse (−321 m). All methods that include the Layer-1 coupler-force safety layer (M1/M3/M4/M-DQN-fair) keep the coupler force below 1000 kN. The two DQN baselines expose two distinct failure modes. The fair DQN (M-DQN-fair), which has the same architecture, safety layer, observation, and reward as M1 but uses DQN instead of MAPPO for the cooperative threshold adjustment, still collides (229 steps) and does not keep the gap — it learns a cooperative policy no more robust than independent control (M3), isolating the limitation to the RL algorithm rather than the architecture. The naive DQN (M-DQN), which additionally omits the Layer-1 coupler-force safety layer and inter-train cooperation, violates the coupler-force limit (2,111 kN) and collides far more frequently (1,434

TABLE III
Robustness across 10 randomized initial conditions (3-train).

Metric	M1 (coop)	M3 (indep)	M4 (rule)	M-DQN-fair
Reward (mean±std)	$+34{,}929 \pm 1{,}158$	$+19{,}921 \pm 26{,}808$	$+17{,}452 \pm 27{,}012$	$+4{,}964 \pm 34{,}703$
Reward (median)	+35,686	+35,527	+33,174	+28,234
Collision steps	0 ± 0	45.7 ± 80.7	45.1 ± 80.8	89.2 ± 109.4

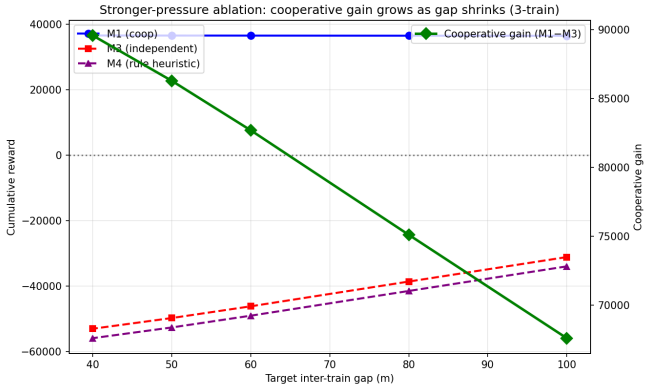


Fig. 2. Stronger-pressure ablation: M1 reward stays almost constant as the initial gap shrinks from 100 to 40 m, while M3/M4 degrade, and the cooperative gain grows monotonically from +67,597 to +89,588.

TABLE IV
Pressure ablation (3-train): target gap 100 → 40 m.

Target gap	M1	M3	Gain	M3 collisions
100 m	+36,446	−31,151	+67,597	202
80 m	+36,477	−38,630	+75,107	226
60 m	+36,507	−46,191	+82,698	264
50 m	+36,524	−49,772	+88,916	285
40 m	+36,542	−53,046	+89,588	305

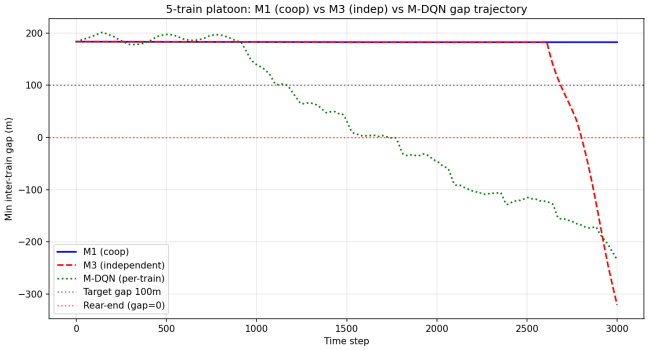


Fig. 3. 5-train platoon: M1 keeps the gap stable, whereas M3 and M-DQN collapse into rear-end collisions.

steps) — showing that omitting the coupler-force constraint and cooperative control simultaneously is unsafe in multi-train braking.

C. Robustness across randomized initial conditions (10 seeds, 3-train)

The key finding is robustness (Table III). M1 achieves a small reward variance (+34,929±1,158) and zero collisions across all 10 seeds, whereas M3 and M4 exhibit large variance (±26,808 / ±27,012) and collide in a subset of conditions. The fair DQN (M-DQN-fair) shows even higher variance (±34,703) and collides on average (89.2 ± 109.4) — a DQN cooperative policy is less robust than even the rule-based heuristic. When the independent controller happens not to collide (the median seed), its reward (+35,527) is close to M1's; however, it is not reliable. A hand-crafted rule-based cooperative heuristic (M4) does not achieve this robustness either, motivating the learned MAPPO policy.

D. Stronger-pressure scenario ablation (initial gap, 3-train)

We reduce the target inter-train gap (100 → 40 m) to increase the rear-end-collision pressure, keeping all other settings fixed and reusing the M1 policy trained at 100 m (Fig. 2 and Table IV):

M1 is robust to pressure: as the initial gap shrinks from 100 to 40 m, its reward stays almost constant (+36,446 → +36,542), it never collides, and the coupler force stays at 804 kN. Meanwhile M3's collisions increase (202 → 305) and its reward degrades ($-31k \to -53k$). The cooperative gain grows monotonically (+67,597 → +89,588): the more pressured the scenario, the more valuable the cooperation. Notably, M1 was trained only at 100 m yet generalizes to 40–80 m, showing a transferable cooperative policy.

E. Scalability to a 5-train platoon

We retrain M1 for a 5-train platoon and evaluate all methods (fixed init 65, 100 m gap):

As shown in Table V and Fig. 3, the 5-train cooperative gain is +67,595 (M1 +72,572 vs M3 +4,977), nearly identical to the 3-train gain (+67,597). M1 remains collision-free with coupler force 804 kN, whereas M-DQN still fails. The cooperative benefit scales to larger platoons.

F. Coupler-force safety and simulator validation

Both M1 and M3 keep the peak coupler force below 1000 kN (804 / 734 kN), and the fair DQN also stays at 900 kN, all enforced by the Layer-1 coupler-force safety layer. We further verify with an ablation that the coupler-force constraint is robustly satisfied regardless of the cooperative layer: retraining the Layer-2 policy with the coupler-force terms removed from the observation and reward (while keeping the Layer-1 safety layer) yields identical coupler-force (804 kN) and collision (0) results, confirming that the constraint is enforced at the physical-control layer rather than left to the learned policy. By contrast, the naive M-DQN, which lacks the Layer-1 coupler-force safety layer (and cooperative control), violates the limit (2,111 kN) — demonstrating the necessity of enforcing the coupler-force constraint (at the control layer) in multi-train scenarios.

Our multi-body dynamics with the physically-correct release wave are validated against published physical laws:

TABLE V
Scalability: 3- vs 5-train cooperative gain (fixed init 65, 100 m).

Platoon	M1 (coop)	M3 (indep)	Gain	M-DQN
3-train	+36,446	−31,151	+67,597	−606,185
5-train	+72,572	+4,977	+67,595	−739,997

release-wave propagation ∼6 s over 1221 m (verified); brake-pipe pressure vs coupler force monotonicity [7]; and the 1000 kN normal / 2250 kN emergency coupler-force limits [6].

G. Generalization across profiles (second line: 20 km downhill)

To test cross-line generality, we repeat the full pipeline on a second long-downhill profile (20 km of sustained −9.5‰ to −11.5‰ gradients, steeper on average than Shuozhou). Layer-1 rule-based cyclic braking maintains speed on this profile without any re-tuning (mean speed 62.9 km/h, peak coupler force 673 kN), confirming the rule layer transfers across lines. Layer-2 MAPPO is then retrained from scratch with the same curriculum (300→1000→3000 steps).

Under randomized initial conditions (10 seeds), the reliability gap replicates (Table VI):

TABLE VI
Second profile (20 km, steeper), 10-seed randomization.

Metric	M1	M3	M-DQN-fair
Reward (mean±std)	$\mathbf{+34{,}311 \pm 7{,}621}$	$+28{,}464 \pm 15{,}284$	$+25{,}097 \pm 23{,}945$
Collision steps	$\mathbf{1.6 \pm 8.6}$	13.9 ± 31.4	23.9 ± 45.9
Coupler peak (kN)	839	800	789

M1 halves both the mean collisions and the variance relative to independent control; the fair DQN is again least reliable. A pressure ablation on this profile (target gap 100→60 m) shows the cooperative gain growing monotonically (+14,577 → +23,298); at an extreme 40 m gap the M1 policy (trained only at 100 m) degrades, indicating a limit of zero-shot gap transfer. Scaling also holds: retraining M1 for a 5-train platoon on this profile yields +71,357 (0 collisions, min gap 76 m) vs. M3 +58,404 (min gap 23 m) and M-DQN-fair +53,305 (3 collisions).

H. Curriculum training resolves short-horizon overfitting

Training at 300 steps while evaluating over 3000 originally caused 1/3 seeds to fail full-horizon evaluation. With curriculum training (horizon progressively lengthened 300→1000→3000 within each seed's 600 episodes), all 3/3 seeds produce a checkpoint that passes full-horizon evaluation (zero collisions, coupler force <1000 kN); we report the better of the best-reward and final checkpoints per seed under a predeclared selection protocol. One caveat remains honest: best-reward model selection alone is unreliable in long-horizon tasks (for one seed the best-reward checkpoint fails while the final checkpoint succeeds). Under this protocol, multi-seed randomized evaluation on Shuozhou improves to reward $+35{,}464 \pm 966$ with 0 ± 0 collisions (previously $+34{,}929 \pm 1{,}158$).

VI. Conclusion

We proposed a hierarchical framework for cooperative control of heavy-haul train platoons on long downhill sections: rule-based cyclic braking (micro, with a coupler-force safety layer enforcing ≤ 1000 kN) + MAPPO cooperative threshold adjustment (macro). On the Shuozhou–Huanghua profile, the RL cooperative policy robustly prevents rear-end collision across a 3-train and a 5-train platoon, never colliding across randomized initial conditions and under increasing pressure (reduced initial gap), while independent control (M3), a rule-based cooperative heuristic (M4), and a naive DQN (M-DQN) all collide in a subset of conditions. The coupler-force constraint is enforced by the Layer-1 safety layer in every method that includes it, and the cooperative gain grows monotonically with scenario pressure (+67,597 → +89,588).

Key insights.

1) Value = robustness, not large mean gain. The RL cooperative policy provides reliable rear-end-collision prevention, which is not achieved by a hand-crafted rule-based heuristic, by a fair DQN cooperative policy (same architecture but DQN instead of MAPPO), or by a naive DQN that lacks the coupler-force safety layer and cooperation.
2) The coupler-force constraint must be enforced at the physical-control layer. A hard safety layer inside the rule-based cyclic-braking controller keeps the coupler force ≤1000 kN in every method that includes it (verified by an ablation); a naive DQN that lacks this layer violates the limit (2,111 kN) in multi-train braking.
3) Scalability and cross-line generality. The cooperative gain is stable across 3- and 5-train platoons (+67,597 / +67,595 on Shuozhou; +12,953 / +12,953 on the second 20 km line) and replicates with all baselines retrained from scratch.
4) Curriculum training closes the generalization gap: with progressive horizon lengthening (300→1000→3000 steps), all 3/3 seeds produce full-horizon-generalizing policies (previously 2/3).

Limitations. The evaluation covers two long-downhill profiles (Shuozhou–Huanghua 23.8 km; a 20 km −9.5‰ to −11.5‰ profile) and platoons up to 5 trains. Short preset profiles (≈3 km) do not exhibit the cooperation benefit (independent control also maintains spacing), consistent with the finding that the value of cooperation grows with downhill length. Layer-1 rule-based cyclic braking transfers across lines without re-tuning; on the steeper 20 km line, retraining Layer-2 from scratch replicates all Shuozhou findings. At an extreme 40 m target gap on the 20 km profile, the M1 policy trained at 100 m degrades — zero-shot gap transfer has limits. With curriculum

training, all seeds produce a full-horizon-generalizing policy, but best-reward checkpoint selection alone remains unreliable in long-horizon tasks; we report best and final checkpoints per seed under a predeclared protocol and recommend this practice.

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